Semantic case studies — corrected programs and changed-but-failing programs

DEEPSEEK · BigCodeBench/937 · α=−5

```
@@ -29,16 +29,5 @@
     return char_counts

-#test_task.py
-import unittest
-from task import task_func
-
-class TestTask(unittest.TestCase):
-    def test_task_func(self):
-        # Test case 1: Basic test
-        input_str = "Hello, World!"
-        expected_output = {'l': 3, 'o': 2, 'h': 1, 'e': 1, 'w': 1,
'r': 1, 'd': 1}
-        self.assertEqual(task_func(input_str), expected_output)
-
-        # Test case 2: Empty string
-        input_str = ""
-        expected_output = {}
+# Test the function
+input_str = "Hello, World!"
+print(task_func(input_str))
```

PASS: removed failing test-module import; solution body unchanged.
Baseline first activation: 40.5% · first code change: 40.5%

DEEPSEEK · BigCodeBench/475 · α=−5

```
@@ -67,5 +67,5 @@

     # Create a histogram of the data
-    ax =
data['dates'].dt.month.value_counts().sort_index().plot(kind='bar')
+    ax =
data['dates'].dt.month.value_counts().sort_index().plot(kind='bar',
color='blue', alpha=0.5)

     # Set the title and labels
@@ -77,7 +77,2 @@

     return ax
-
-#tests/test_task_func.py
-import unittest
-import pandas as pd
-from task_func import task_func
```

FAIL: test imports removed, but evaluator still reports missing locale.
Baseline first activation: 93.0% · first code change: 77.8%

CODELLAMA · BigCodeBench/490 · α=−5

```
@@ -26,7 +26,10 @@
     'Emma'
     """
-    return xmltodict.parse(s)
+    result = xmltodict.parse(s)
+    with open(file_path, 'w') as f:
+        json.dump(result, f)
+    return result

-def task_func2(s, file_path):
[Additional duplicate task_func_2 omitted from this excerpt.]
```

PASS: file-writing behavior restored; duplicate helper remains in output.
Baseline first activation: 2.0% · first code change: 0.2%

CODELLAMA · BigCodeBench/878 · α=−5

```
@@ -56,30 +56,22 @@
     if target is None or target not in data:
         raise ValueError("The target column name is not in the
DataFrame.")
+    if test_size is None or test_size < 0 or test_size > 1:
+        raise ValueError("The fractional size of the test set is
not between 0 and 1.")
     if random_state is None:
-        random_state = None
-    if test_size is None:
-        test_size = 0.2
-    if test_size > 1:
-        raise ValueError("The fractional size of the test set is
greater than 1.")
-    if test_size < 0:
-        raise ValueError("The fractional size of the test set is
less than 0.")
-    if test_size < 0.01:
-        raise ValueError("The fractional size of the test set is
less than 0.01.")
-    if test_size > 1:
-        raise ValueError("The fractional size of the test set is
greater than 1.")
-    if test_size < 0:
[Excerpt ends here; complete diff in CASE_INSPECTION.md]
```

FAIL: validation changes, but repeated checks remain and the task still fails.
Baseline first activation: 7.6% · first code change: 14.4%